

Effect of a Change in the Eligibility Criteria for Multidisciplinary Kidney Care

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Key Points

- Multidisciplinary kidney care eligibility changed from eGFR ≤ 33.4 ml/min per 1.73 m^2 to 2-year kidney failure risk $\geq 10\%$ (includes eGFR and proteinuria).
- After the change, we found no difference in outcomes such as dialysis initiation, preparation for dialysis, and hospitalizations.
- The results of this study help guide decision making with respect to the care of patients with advanced CKD.

Abstract

Background We examined the effect of a change in the eligibility criteria for multidisciplinary kidney care implemented across Ontario, Canada, on April 1, 2016 (criteria moved from an eGFR ≤ 33.4 ml/min per 1.73 m^2 to 2-year kidney failure risk $\geq 10\%$, which incorporates assessment of proteinuria, or an eGFR < 15).

Methods Population-based, interrupted time series analysis using administrative health care databases that included adults with an outpatient eGFR ≤ 33.4 ml/min per 1.73 m^2 followed by a nephrologist ($N=97,299$). We examined multidisciplinary clinic and nephrologist visits, dialysis-related outcomes, hospital encounters, and mortality by monthly interval between October 1, 2013, and February 1, 2020. Autoregressive integrated moving average models were tested for level (immediate) and slope (over time) changes in outcomes after the intervention and were fit on preintervention change data for projected trends.

Results After intervention, there was a significant monthly decline in the multidisciplinary clinic visit rate of 1.49 visits per 100 person-years (95% confidence interval [CI], -2.83 to -0.14) and a significant monthly increase in the proportion of patients with a referral to multidisciplinary care within 1 year before starting dialysis (*i.e.*, late referral; positive slope change, 0.67%; 95% CI, 0.06 to 1.28). The intervention was not associated with significant changes in the proportion of patients initiating dialysis, initiating dialysis with a central venous catheter or during a hospitalization, using home dialysis, or in the rate of nephrologist visits. Immediately after intervention, there were small, significant increases in mortality and hospital encounters, but importantly, there was no monthly (slope) change in mortality and a significant negative monthly change in hospital encounters (-0.24 encounters per 100 person-years; 95% CI, -0.38 to -0.09).

Conclusions Changing the eligibility criteria for multidisciplinary kidney care from an eGFR ≤ 33.4 ml/min per 1.73 m^2 to a 2-year kidney failure risk $\geq 10\%$ or an eGFR < 15 ml/min per 1.73 m^2 resulted in significantly fewer patients receiving such care without negatively affecting important clinical outcomes.

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Introduction

Owing to complex care needs, patients with advanced CKD benefit from care in multidisciplinary kidney clinics, with access to a team of interdisciplinary health care professionals (*i.e.*, nephrologist, nurse, pharmacist, dietitian, and social worker). These clinics focus on slowing CKD progression, treating complications, and preparing patients for dialysis, kidney transplant, or conservative kidney care.¹ Receipt of multidisciplinary kidney care is associated with a better dialysis preparation and lower rates of hospitalization and mortality.^{2–8} Most Canadian centers, and some in the United States, have implemented multidisciplinary kidney clinics, and guidelines recommend that patients with advanced CKD receive multidisciplinary care.^{8–11}

Historically, the eligibility criterion for multidisciplinary kidney clinic care in Ontario was an eGFR ≤ 33.4 ml/min per 1.73 m² (on the basis of the eGFR cutoff for CKD category 4 of <30 ml/min per 1.73 m² with a slight increase to account for prior lack of universal standardization of creatinine assays). However, some patients with a low eGFR do not have a high risk of progressing to kidney failure. In particular, many older patients with an eGFR <30 ml/min per 1.73 m² have no or very low proteinuria (*i.e.*, CKD category G4A1) and therefore a very low risk of kidney failure ($>30\%$ of patients age 65–74 years, approximately 50% of patients age 75–84 years, and approximately 70% of patients age ≥ 85 years).^{12–16} Multidisciplinary kidney care is costly; annual funding for a patient receiving multidisciplinary care in Ontario compared with general kidney clinic care is 6.6 times higher.¹⁷

To use finite resources more efficiently, the eligibility criteria for multidisciplinary kidney care were revised in Ontario, Canada, in April 2016, moving from a solely eGFR-based criterion to using the four-variable kidney failure risk equation (KFRE).¹² Specifically, people were eligible for multidisciplinary kidney care if they had a $\geq 10\%$ 2-year predicted risk of kidney failure or an eGFR <15 ml/min per 1.73 m². This policy change created a natural experiment to study the change in eligibility criteria on patient outcomes. We hypothesized that this change would lead to a reduction in the number of patients followed in multidisciplinary kidney clinics without negatively affecting patient outcomes.

Methods

Study Design and Setting

We conducted a retrospective population-based, interrupted time series analysis with administrative health care databases linked using unique encoded identifiers and analyzed at ICES (ices.on.ca/) in Ontario, Canada. All Ontario residents (population of more than 16 million people) receive universal access to hospital and physician services, and older residents (65 years and older) receive coverage for prescription medications. Coverage for medical services and medications from a single provincial payer provide a comprehensive set of health administrative data. Kidney care in Ontario is organized across 27 regional programs, with each program having its own Multi-Care Kidney Clinic. Patients can be referred to Multi-Care Kidney Clinics only by nephrologists.

On April 1, 2016, the eligibility criteria for Multi-Care Kidney Clinics in Ontario, Canada, were changed from an eGFR ≤ 33.4 ml/min per 1.73 m² to a lower level of kidney function (eGFR <15 ml/min per 1.73 m²) or an estimated 2-year risk of kidney failure $\geq 10\%$ as determined by the four-variable KFRE.¹² For patients who were already followed in Multi-Care Kidney Clinics before April 1, 2016, the new eligibility criteria were applied using a staged approach. In the first year (April 1, 2016–March 31, 2017), prevalent Multi-Care Kidney Clinic patients could continue to receive funding for multidisciplinary care if their KFRE was $\geq 5\%$. In the second year (April 1, 2017–March 31, 2018), the funding threshold for prevalent patients increased to $\geq 7.5\%$, and in the third year (April 1, 2018–March 31, 2019), all prevalent patients had to have a KFRE $\geq 10\%$ or an eGFR <15 to receive funding. From April 1, 2016 onward, incident Multi-Care Kidney Clinic patients (*i.e.*, new referrals) had to meet the new eligibility criteria to receive funding. Programs could still choose to maintain ineligible patients in Multi-Care Kidney Clinics, but they would not receive funding for such patients.

The study was conducted according to a prespecified protocol. The use of data in this project was authorized under section 45 of Ontario's Personal Health Information Protection Act, which did not require review by a Research Ethics Board. The reporting of this study follows the Reporting of studies Conducted using Observational Routinely collected health Data (RECORD) guidelines for observational studies (Supplemental Appendix A).¹⁸

Data Sources

Patients meeting both old and new Multi-Care Kidney Clinic eligibility criteria were identified using the Ontario Laboratories Information System. The Ontario Renal Reporting System was used to determine Multi-Care Kidney Clinic follow-up. Full details on all databases and codes used can be found in Supplemental Appendices B and C.

Study Cohort

The study was completed at the population level. We divided each study year (*i.e.*, fiscal year) into months (12-month intervals/year) between October 1, 2013, and February 1, 2020 (77 intervals total). We selected this end date to minimize the effect of the coronavirus disease 2019 pandemic (restrictions on in-person clinical activities began March 2020). We identified adults (≥ 18 years) with most recent outpatient eGFR ≤ 33.4 ml/min per 1.73 m² (*i.e.*, eligibility criteria before April 1, 2016; eGFR determined using the 2009 CKD Epidemiology Collaboration formula¹⁹ in the year before each monthly interval). The index date (start of follow-up) was the first day of each monthly interval. We then restricted the cohort to patients receiving ongoing care by a nephrologist defined by evidence of at least two outpatient nephrologist visits separated by ≥ 6 months in the past 2 years, where at least one visit was within the 15 months before the index date. We excluded patients with any history of dialysis or a kidney transplant. Although each patient could be included only once per monthly interval, a patient could be in more than one interval if they still met all inclusion criteria before the index date.

Quality of Care Indicators and Outcomes

Count outcomes (*i.e.*, recurrent) were reported as rates per 100 person-years (p-y) and binary outcomes were reported as proportions. For detailed outcome definitions, refer to [Supplemental Appendix D](#). We examined the following outcomes by monthly interval: (1) proportion of prevalent Multi-Care Kidney Clinic patients; (2) rate of Multi-Care Kidney Clinic visits; (3) rate of nephrologist visits; (4) proportion initiating dialysis; (5) proportion, among those who started dialysis, with late referral for Multi-Care Kidney Clinic care (defined by receipt of care for <1 year before dialysis initiation); (6) proportion, among those who started dialysis, with a nondeferred dialysis start (defined by an eGFR >9.5 ml/min per 1.73 m² at dialysis start as per the Ontario Renal Network; (7) proportion with dialysis initiation during a hospitalization or with a central venous catheter (CVC); (8) proportion with home dialysis use; (9) rate of all-cause hospital encounters (admissions or emergency department visits); (10) rate of hospital encounters with a cardiovascular complication; and (11) proportion with all-cause mortality. Within each monthly interval, outcomes 2 to 9 were assessed among all patients, with censoring for dialysis initiation, preemptive kidney transplantation, or death. For outcome 1, this was defined by evidence of a Multi-Care Kidney Clinic visit within 1 year before the index date. Owing to a low incidence, preemptive kidney transplant could not be examined as an outcome.

Statistical Analysis

We reported baseline characteristics for the whole cohort (patients with an eGFR ≤33.4 ml/min per 1.73 m² receiving nephrology care) and for prevalent Multi-Care Kidney Clinic patients for the monthly intervals of October 2013, March 2016, March 2018, and February 2020. We also compared characteristics for those who received Multi-Care Kidney Clinic care with those who did not over the same four monthly intervals using standardized differences. Standardized differences ≥10% were considered a significant difference between categories.²⁰ Analyses were not performed by Multi-Care Kidney Clinic status but rather for the total cohort of patients meeting our inclusion criteria. For each monthly interval, we examined indicators and outcomes descriptively. We reported the mean of the monthly rate or proportion for each outcome during the period before the policy change and after the change, and the difference in means between these periods.

We then conducted an interrupted time series analysis using autoregressive integrated moving average (ARIMA) models with step and ramp functions (further details in [Supplemental Appendix E](#)). The step coefficient represents the immediate and sustained difference in the outcome observed in the first month after the policy took effect on April 1, 2016, relative to the value that would have been expected had the preintervention trend continued. The ramp coefficient quantifies the month-to-month change in the postintervention slope compared with the preintervention slope. All analyses were completed using the full cohort and then repeated using the subgroups of patients who met the new Multi-Care Kidney Clinic eligibility criteria (2-year kidney failure risk ≥10% or eGFR <15, referred to as high kidney failure risk) versus did not meet the new eligibility criteria (referred to as low kidney failure

risk). For patients with a missing urine albumin-to-creatinine ratio, those with an eGFR >15 ml/min per 1.73 m² were classified as low kidney failure risk and those with an eGFR <15 ml/min per 1.73 m² were classified as high kidney failure risk. Two-sided *P* values < 0.05 were considered significant. All analyses were completed using SAS version 9.4 (SAS Institute, Cary, NC).

Results

Baseline Characteristics

Cohort selection is detailed in [Supplemental Figure 1](#). Our final cohort included 97,299 patients with an eGFR ≤33.4 ml/min per 1.73 m² followed by a nephrologist, with each monthly interval including a minimum and maximum of 25,650 patients (October 2013) and 40,041 patients (August 2019), respectively ([Supplemental Figure 2](#)). Patient characteristics for the full cohort and for Multi-Care Kidney Clinic versus not, across selected years, are presented in [Supplemental Tables 1 and 2](#). Non-Multi-Care Kidney Clinic patients compared with Multi-Care Kidney Clinic patients were older, were more likely to be female, and had a lower 2-year risk of kidney failure; comorbidities were similar. Multi-Care Kidney Clinic patients in February 2020 (postintervention) compared with March 2016 (preintervention) were younger (72 years [SD 13] versus 74 years [SD 14]), less likely to be female (40% versus 46%), and had a higher 2-year kidney failure risk (mean 35% [SD 25] versus 25% [SD 25]); comorbidities were similar ([Table 1](#)).

Indicators and Outcomes by Monthly Interval for the Full Cohort

The proportion of the cohort followed in a Multi-Care Kidney Clinic peaked just before the intervention (*n*=11,946 of 31,026, 39%, March 2016) and then gradually declined to a plateau by 2018 (*n*=10,423 of 39,404, 27%, June 2018) ([Supplemental Figure 3](#)). Results for the ARIMA modeling are presented in [Table 2](#). There was a significant monthly decline in the Multi-Care Kidney Clinic visit rate of 1.49 visits per 100 p-y after the intervention (95% confidence interval [CI], −2.83 to −0.14) ([Figure 1](#)). There was a significant monthly rise (positive slope) after the intervention in late referrals for multidisciplinary care (ramp estimate=0.67%; 95% CI, 0.06 to 1.28) ([Figure 2](#)). Compared with preintervention, there was a significant immediate (step) decline postintervention in the proportion of patients with a nondeferred dialysis start (step estimate=−6.18%; 95% CI, −8.45 to −3.92), but there was no significant monthly change (slope) postintervention ([Supplemental Figure 4](#)). The intervention was not associated with significant changes for several outcomes, including the rate of nephrologist visits ([Supplemental Figure 5](#)), the proportion with dialysis initiation ([Supplemental Figure 6](#)), the proportion initiating dialysis with a CVC or during a hospitalization ([Supplemental Figure 7](#)), the proportion with home dialysis use ([Supplemental Figure 8](#)), and the rate of hospital encounters with a cardiovascular complication ([Supplemental Figure 9](#)). Compared with preintervention, there were significant immediate (step) increases in both all-cause hospital encounters (step estimate=8.71 encounters/100 p-y; 95% CI, 4.25 to

Table 1. Baseline characteristics for prevalent Multi-Care Kidney Clinic patients in Ontario, Canada, for selected years

Characteristics	Description	October 1, 2013	March 1, 2016	March 1, 2018	February 1, 2020
		N=6637	N=11,946	N=10,613	N=9942
Age, yr	Mean (SD)	73 (13)	74 (13)	72 (14)	72 (14)
Sex, N (%)	Female	3003 (45)	5531 (46)	4442 (42)	4016 (40)
Rurality	Rural	692 (10)	1236 (10)	1115 (11)	1007 (10)
	Urban	5945 (90)	10,685 (89)	9473 (89)	8909 (90)
	Missing	0 (0)	25 (0.2)	25 (0.2)	26 (0.3)
	Income quintile, N (%)				
	1 (low)	1473 (22)	3063 (26)	2648 (25)	2657 (27)
	2	1456 (22)	2736 (23)	2443 (23)	2260 (23)
	3	1282 (19)	2361 (20)	2059 (19)	1809 (18)
	4	1273 (19)	1943 (16)	1853 (18)	1691 (17)
	5 (high)	1129 (17)	1811 (15)	1582 (15)	1495 (15)
	Missing	24 (0.4)	32 (0.3)	28 (0.3)	30 (0.3)
Serum creatinine, mg/dl	Mean (SD)	3.1 (1.2)	3.0 (1.2)	3.3 (1.3)	3.4 (1.2)
eGFR, ml/min per 1.73 m ²	Mean (SD)	19 (7)	20 (7)	18 (6)	18 (6)
Urine ACR test, mg/g, N (%)	<26.5	574 (9)	1152 (10)	440 (4)	258 (3)
	26.5–265.2	1680 (25)	3273 (27)	2848 (27)	2439 (25)
	>265.2	2373 (36)	5227 (44)	6720 (63)	6879 (69)
	Not detectable	290 (4)	580 (5)	118 (1%)	83 (0.8)
	Missing/test not done	1720 (26)	1714 (14)	487 (5)	283 (3)
Serum potassium, mEq/L	Mean (SD)	4.6 (0.5)	4.6 (0.5)	4.7 (0.6)	4.7 (0.6)
	Missing/test not done, N (%)	85 (1)	114 (1)	31 (0.3)	19 (0.2)
Hemoglobin, g/dl	Mean (SD)	11.4 (1.5)	11.5 (1.6)	11.4 (1.7)	11.3 (1.7)
	Missing/test not done, N (%)	142 (2)	63 (0.5)	27 (0.3)	40 (0.4)
PTH, pg/ml	Median (IQR)	118 (74–186)	122 (76–198)	133 (82–216)	143 (89–236)
	Missing/test not done, N (%)	2147 (32)	2714 (23)	1834 (17)	1680 (17)
2-yr KFRE, % ^a	Mean (SD)	30 (30)	25 (25)	30 (30)	35 (25)
High risk of kidney failure (2-yr risk ≥10% or eGFR <15) ^b	N (%)	3625 (55)	6694 (56)	8601 (81)	8625 (87)
No. of previous emergency room visits	Mean (SD)	1.2 (1.9)	1.2 (2.0)	1.2 (2.1)	1.2 (2.1)
No. of previous primary care physician visits	Mean (SD)	11.2 (11.6)	10.3 (11.5)	10.2 (11.9)	9.7 (11.8)
No. of visits to a nephrologist	Mean (SD)	3.6 (2.0)	3.4 (1.9)	3.6 (2.0)	3.5 (1.9)
Charlson Comorbidity score, N (%)	0	292 (4)	516 (4)	387 (4)	371 (4)
	1	154 (2)	293 (3)	240 (2)	216 (2)
	2	4654 (70)	8562 (72)	7656 (72)	7240 (73)
	3+	1537 (23)	2575 (22)	2330 (22)	2115 (21)
Diabetes (any time prior)	N (%)	4082 (62)	7426 (62)	6922 (65)	6566 (66)
Hypertension (any time prior)	N (%)	6157 (93)	11,043 (92)	9724 (92)	9099 (92)
Myocardial infarction	N (%)	502 (8)	893 (8)	742 (7)	670 (7)
Congestive heart failure	N (%)	2149 (32)	3754 (31)	3192 (30)	3025 (30)
Coronary artery disease	N (%)	2409 (36)	4125 (35)	3450 (33)	3190 (32)
Peripheral vascular disease	N (%)	262 (4)	403 (3)	348 (3)	315 (3)
Stroke, excluding TIA	N (%)	213 (3)	398 (3)	381 (4)	362 (4)
Atrial fibrillation/flutter	N (%)	1870 (28)	3513 (29)	2930 (28)	2825 (28)
Ventricular arrhythmia	N (%)	51 (0.8)	105 (0.9)	86 (0.8)	68 (0.7)
Coronary artery bypass graft surgery	N (%)	137 (3)	255 (2)	231 (2)	214 (2)
Percutaneous coronary intervention	N (%)	288 (4)	591 (5)	499 (5)	487 (5)
COPD	N (%)	1995 (30)	3519 (30)	2970 (28)	2709 (27)
Chronic liver disease, including cirrhosis	N (%)	369 (6)	641 (5)	674 (6)	667 (7)
Major cancer	N (%)	988 (15)	1919 (16)	1673 (16)	1567 (16)
No. of patients with available medication data (66+ on index date); used as denominator for medications below	N (%)	4952 (75)	9223 (77)	7669 (72)	7152 (72)
Diabetes drugs	N (%)	2252 (46)	4184 (45)	3827 (50)	3777 (53)
SGLT2 inhibitors	N (%)	0 (0)	11 (0.1)	34 (0.4)	171 (2)
No. of antihypertensive meds, N (%)	0	223 (5)	597 (7)	501 (7)	391 (6)
	1	294 (6)	663 (7)	517 (7)	430 (6)
	2	512 (10)	1039 (11)	827 (11)	727 (10)
	3	626 (13)	1101 (12)	860 (11)	780 (11)
	4	595 (12)	1082 (12)	884 (12)	823 (12)
	5+	2702 (55)	4741 (51)	4080 (53)	4001 (56)
ACE inhibitors	N (%)	1250 (25)	2304 (25)	1690 (22)	1539 (22)
ARBs	N (%)	1617 (33)	2706 (29)	2266 (30)	2141 (30)

Table 1. Continued

Characteristics	Description	October 1, 2013	March 1, 2016	March 1, 2018	February 1, 2020
		N=6637	N=11,946	N=10,613	N=9942
ACE inhibitors or ARBs	N (%)	2804 (57)	4950 (54)	3925 (51)	3648 (51)
Statins	N (%)	3592 (73)	6733 (73)	5660 (74)	5378 (75)
β blockers	N (%)	2712 (55)	4744 (51)	3951 (52)	3759 (53)
Calcium channel blockers	N (%)	3093 (63)	5472 (59)	4998 (65)	4792 (67)
K-sparing diuretic	N (%)	352 (7)	558 (6)	431 (6)	374 (5)
Thiazide diuretic	N (%)	1096 (22)	1953 (21)	1639 (21)	1519 (21)
Loop diuretic	N (%)	2717 (55)	4620 (50)	3785 (49)	3564 (50)
α blocker	N (%)	920 (19)	1520 (17)	1581 (21)	1627 (23)
Centrally acting antiadrenergic	N (%)	107 (2)	173 (2)	154 (2)	126 (2)

ACE, angiotensin-converting enzyme; ACR, albumin-to-creatinine ratio; ARB, angiotensin receptor blockers; COPD, chronic obstructive pulmonary disease; IQR, interquartile range; KFRE, kidney failure risk equation; PTH, parathyroid hormone; SGLT2, sodium-glucose cotransporter 2; TIA, transient ischemic attack.

^aThose with missing urine albumin-to-creatinine ratio were excluded from this value.

^bThose with missing urine albumin-to-creatinine ratio were categorized as high kidney failure risk if their eGFR was <15 and low kidney failure risk if their eGFR was >15.

13.17) and all-cause mortality (step estimate=0.038%; 95% CI, 0.004 to 0.072) after the intervention. However, post-intervention, there was no significant monthly change (slope) for mortality (Figure 3), and there was a significant monthly decline (negative slope) for all-cause hospital encounters (ramp estimate=−0.24 hospital encounters per 100 p-y; 95% CI, −0.38 to −0.09) (Figure 4).

Indicators and Outcomes by Monthly Interval for Those with High versus Low Kidney Failure Risk

In patients with high kidney failure risk, the proportion followed in a Multi-Care kidney Clinic gradually increased before the intervention, peaked shortly thereafter (61%, November 2016), and then a plateau was reached by the summer of 2017 (58%, July 2017; Supplemental Figure 3). In patients with low kidney failure risk (*i.e.*, not meeting the new eligibility criteria), the proportion followed in a Multi-Care Kidney Clinic peaked before the intervention (26%, March 2016) and then steadily declined (6%, December 2019; Supplemental Figure 3). The ARIMA modeling for patients with high versus low kidney failure risk is presented in Tables 3 and 4, respectively. After the intervention, there was a monthly decline in the Multi-Care Kidney Clinic visit rate for both the high- and low-kidney failure risk groups, but the decline was greater for those with low kidney failure risk (ramp estimate=−1.43 visits per 100 p-y; 95% CI, −2.18 to −0.66 for high risk, and −2.10 visits per 100 p-y; 95% CI, −4.20 to −0.00 for low risk) (Figure 1). There was a similar, significant monthly decline in the nephrologist visit rate for both high- and low-kidney failure risk groups after the intervention (ramp estimate=−0.59 visits per 100 p-y; 95% CI, −1.08 to −0.11 for high risk, and −0.50 visits per 100 p-y; 95% CI, −0.96 to −0.03 for low risk; Supplemental Figure 5). There was an immediate (step) decline but no difference in slope for dialysis initiation (Supplemental Figure 6) and dialysis initiation during a hospitalization or with a CVC (Supplemental Figure 7) for both the high- and low-risk groups. After the intervention, there was a significant monthly increase (positive slope) in late

referrals for multidisciplinary care in both the high-risk and low-risk groups (ramp estimate=0.70%; 95% CI, 0.04 to 1.35 for high risk, and 0.27%; 95% CI, 0.12 to 0.43 for low risk) (Figure 2). Postintervention, there was a significant immediate (step) decline in nondeferred dialysis starts in high-risk patients (step estimate=−5.45%; 95% CI, −7.73 to −3.10), but the slope did not change after intervention; there were no significant changes for this outcome postintervention in low-risk patients (Supplemental Figure 4). Postintervention, there was a significant monthly decline (negative slope) in home dialysis initiation in high-risk patients (ramp estimate=−0.002%; 95% CI, −0.004 to −0.000); in low-risk patients, a significant immediate decline (step estimate=−0.003%; 95% CI, −0.005 to −0.001) but no difference in slope (ramp estimate) was observed (Supplemental Figure 8). No immediate or slope changes in mortality were observed for high-risk patients. Postintervention, there was a significant immediate increase in mortality for low-risk patients (step estimate=0.04%; 95% CI, 0.01 to 0.07) but no difference in the slope (Figure 3). In high-risk patients, no significant immediate or slope changes were observed for all-cause hospital encounters. For low-risk patients, there was a significant immediate increase in hospital encounters (step estimate=8.90 encounters per 100 p-y; 95% CI, 4.70 to 13.09), but a significant monthly decline (negative slope) after intervention (ramp estimate=−0.34 per 100 p-y; 95% CI, −0.47 to −0.20) (Figure 4).

Discussion

In this retrospective time series study that included 97,299 patients with CKD followed by a nephrologist, we found that a change from eGFR-based to kidney failure risk-based criteria for receipt of multidisciplinary kidney care resulted in a significant decline in the number of patients receiving this resource intensive care. Preintervention in March 2016, there were 11,946 of 31,026 patients (39%) receiving Multi-Care Kidney Clinic care, which gradually declined postintervention to a plateau by June 2018 to

Table 2. Difference in clinical indicators and outcomes after a change in the eligibility criteria for Multi-Care Kidney Clinics

Outcome	Pre-Multi-Care Kidney Clinic Eligibility Change		Post-Multi-Care Kidney Clinic Eligibility Change		Difference in Means from Post- versus Pre-Multi- Care Kidney Clinic Eligibility Change	ARIMA Model (p, d, q)	April 1, 2016, Multi-Care Kidney Clinic First Eligibility Criteria Change Point	
	Total # of Events	Mean of the Monthly Rate or Proportion (SD)	Total # of Events	Mean of the Monthly Rate or Proportion (SD)			Step Function ^a	Ramp Function ^b
							Estimate (95% CI)	Estimate (95% CI)
Multi-Care Kidney Clinic visit ^c	82,995	120.4 (12.1)	138,309	99.5 (12.8)	−20.8	(0, 1, 1) (0, 12, 0)	0.08 (−6.83 to 7.00)	−1.49 (−2.83 to −0.14)
Nephrologist visit ^c	199,459	290.3 (26.2)	379,965	272.1 (25.8)	−18.2	(3, 0, 0) (0, 12, 0)	1.06 (−8.18 to 10.30)	−0.43 (−0.92 to 0.06)
Dialysis initiation	4512	0.5% (0.06)	8524	0.5% (0.05)	0.0%	(0, 1, 1)	−0.02 (−0.09 to 0.04)	−0.000 (−0.005 to 0.005)
Late referral for multidisciplinary kidney care ^d	2428	54.1% (7.8)	3780	44.4% (3.3)	−9.7%	(3, 1, 0)	−3.57 (−8.50 to 1.37)	0.67 (0.06 to 1.28)
Nondeferred dialysis starts ^d	2256	49.9% (3.9)	3714	43.6% (3.4)	−6.3%	(1, 0, 0)	−6.18 (−8.45 to −3.92)	−0.01 (−0.08 to 0.06)
Initiating dialysis with a CVC or during a hospitalization	2753	0.3% (0.05)	5076	0.3% (0.04)	0.0%	(2, 1, 0)	−0.01 (−0.09 to 0.07)	−0.000 (−0.011 to 0.012)
Home dialysis	1113	0.1% (0.02)	2232	0.1% (0.03)	0.0%	(0, 0, 1)	0.01 (−0.01 to 0.03)	−0.001 (−0.001 to 0.000)
All-cause hospital encounters ^c	94,514	137.1 (5.7)	196,929	140.7 (4.9)	3.6	(1, 0, 0)	8.71 (4.25 to 13.17)	−0.24 (−0.38 to −0.09)
Hospital encounter with a cardiovascular complication ^c	32,844	47.8 (2.4)	63,802	45.6 (2.5)	−2.2	(0, 0, 0) (12, 12, 0)	−1.01 (−2.11 to 0.09)	−0.03 (−0.07 to 0.01)
All-cause mortality	5494	0.6% (0.08)	12,684	0.7% (0.08)	0.1%	(0, 0, 0) (12, 12, 0)	0.038 (0.004 to 0.072)	−0.000 (−0.002 to 0.001)

ARIMA, autoregressive integrated moving average; CI, confidence interval; CVC, central venous catheter.

^aThe step coefficient represents the immediate and sustained difference in the outcome observed in the first month after the policy took effect on April 1, 2016, relative to the value that would have been expected had the preintervention trend continued unchanged.

^bThe ramp coefficient quantifies the month-to-month change in the postintervention slope compared with the preintervention slope.

^cRate per 100 person-years.

^dDenominator patients initiating dialysis.

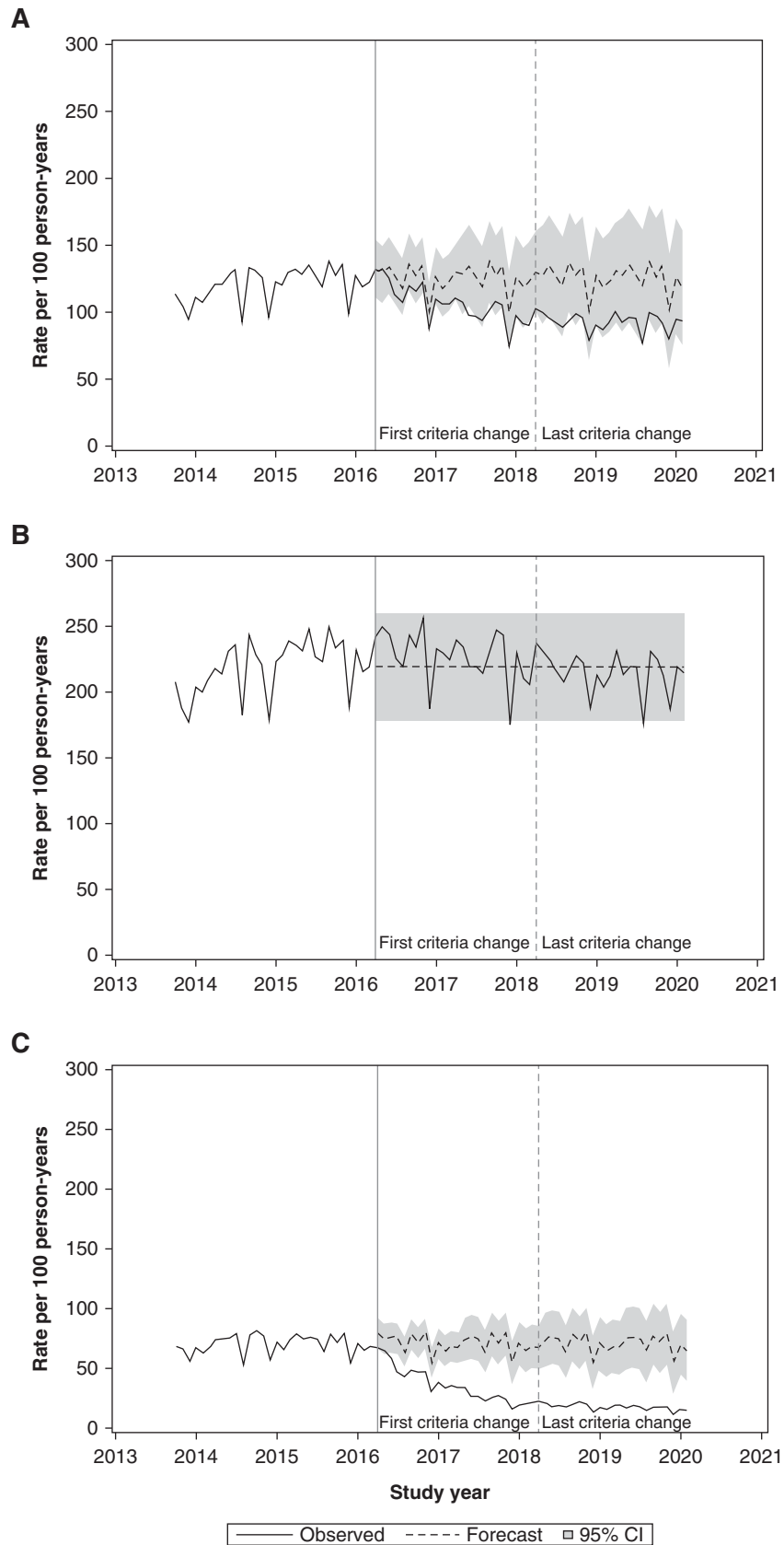


Figure 1. Multi-Care Kidney Clinic visit rates from 2013 to 2020 (per 100 p-y). (A) Overall cohort. (B) High-risk kidney failure group. (C) Low-risk kidney failure group. CI, confidence interval; p-y, person-years.

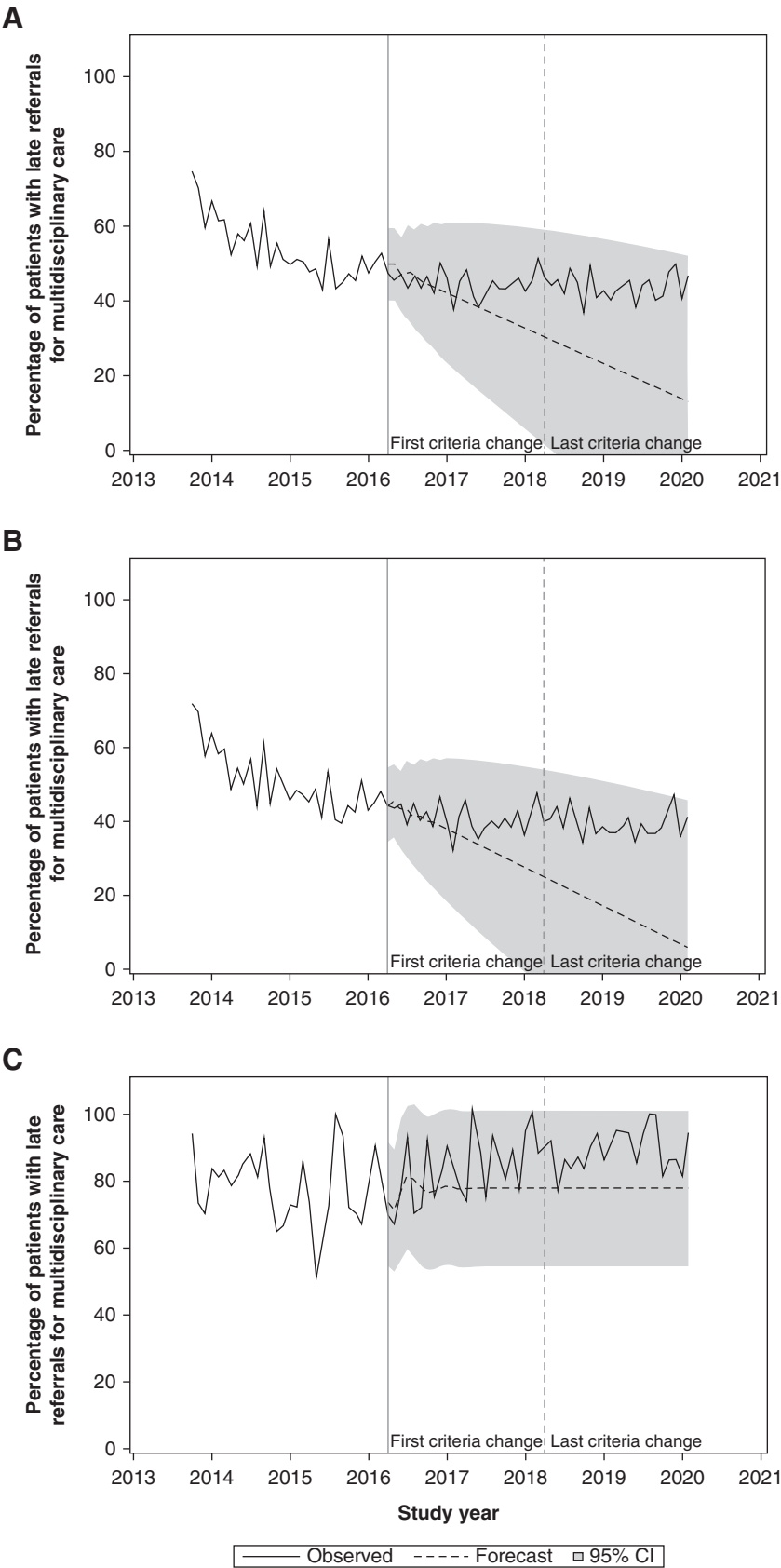


Figure 2. Late referral for multidisciplinary care from 2013 to 2020 (%). (A) Overall cohort. (B) High-risk kidney failure group. (C) Low-risk kidney failure group. Denominator, patients who started dialysis. Late referral defined as <1 year before dialysis initiation.

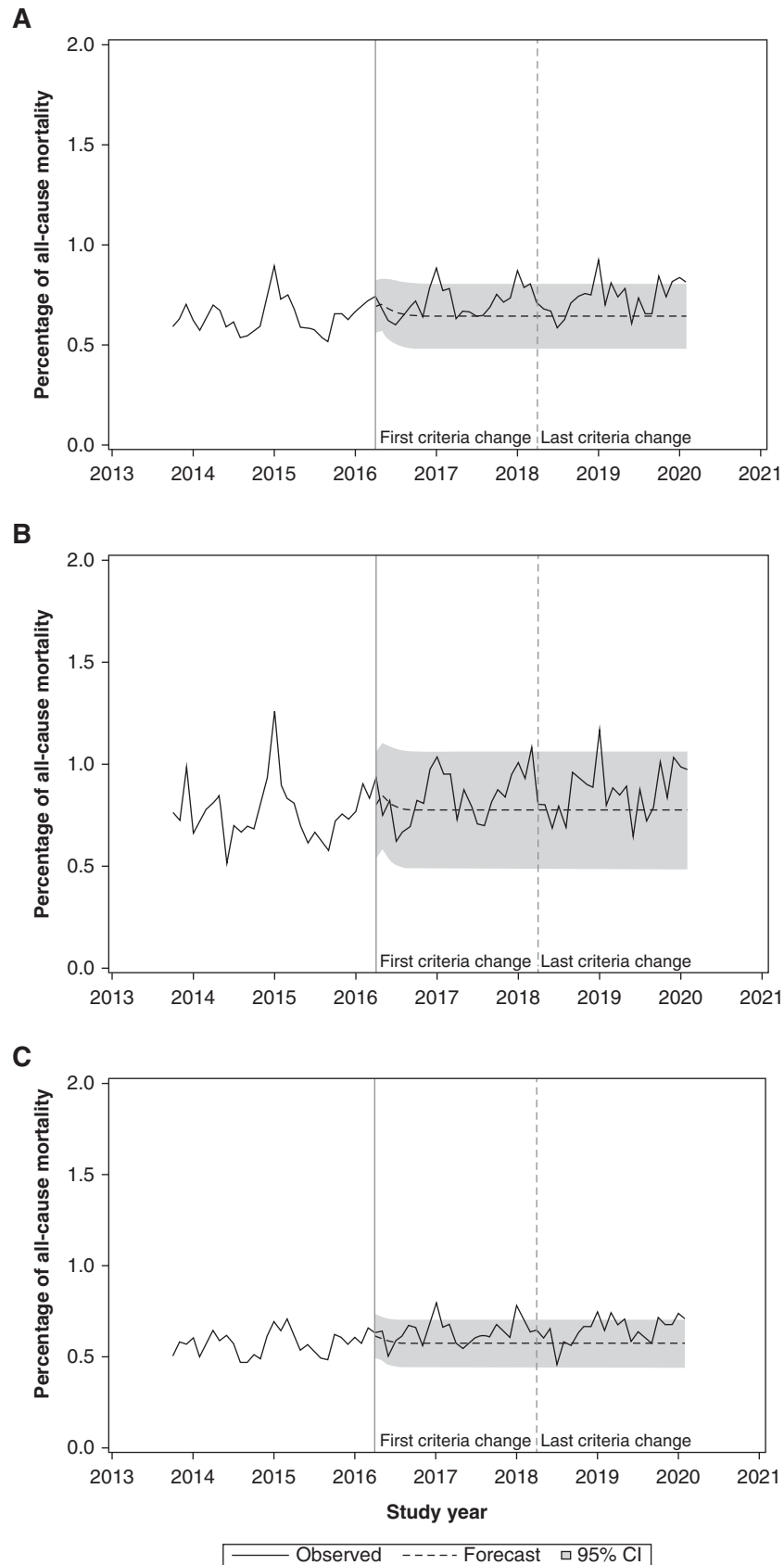


Figure 3. All-cause mortality from 2013 to 2020 (%). (A) Overall cohort. (B) High-risk kidney failure group. (C) Low-risk kidney failure group.

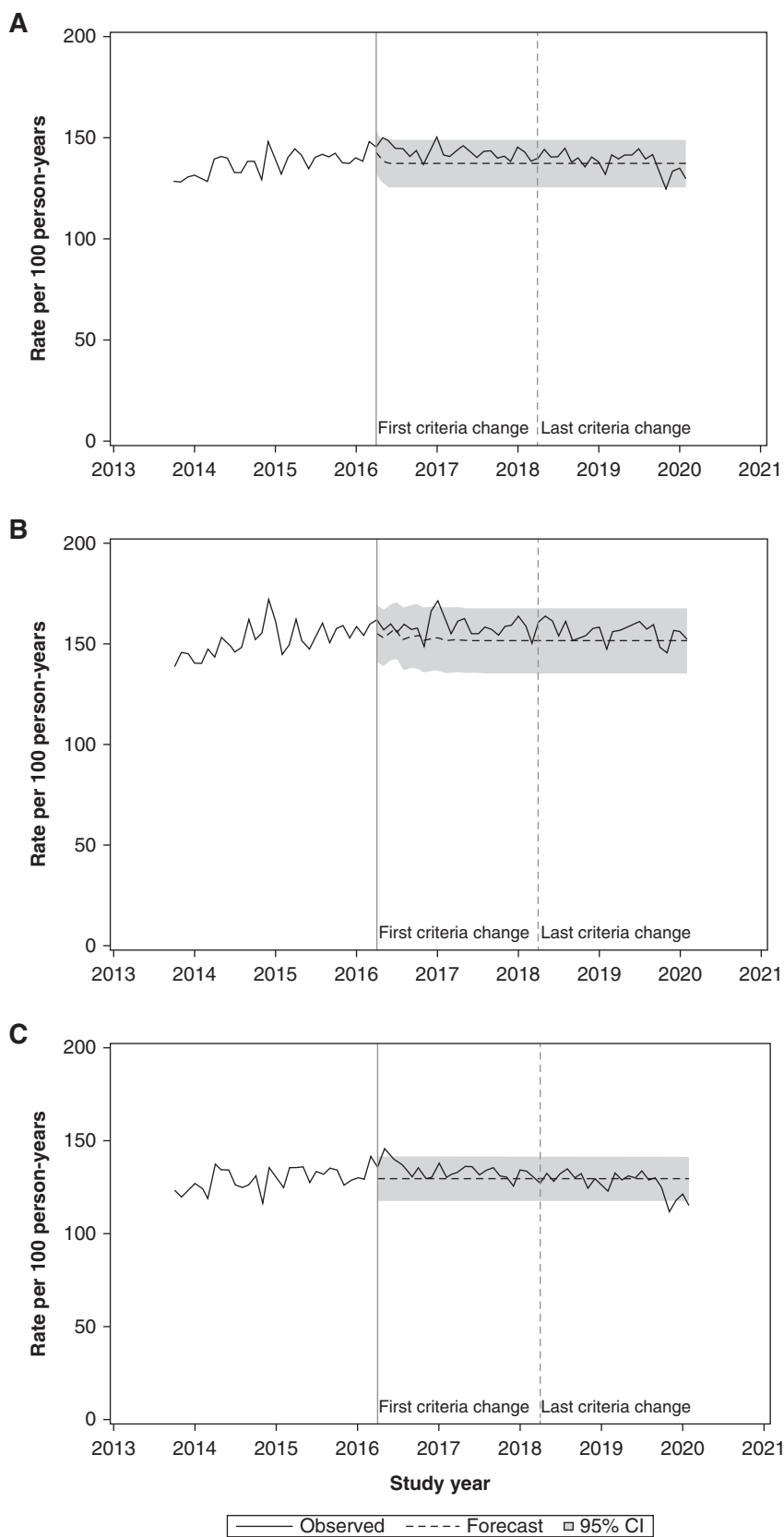


Figure 4. All-cause hospital encounter rate from 2013 to 2020 (per 100 p-y). (A) Overall cohort. (B) High-risk kidney failure group. (C) Low-risk kidney failure group.

Table 3. Difference in clinical indicators and outcomes after a change in the eligibility criteria for Multi-Care Kidney Clinics for individuals in the high-risk kidney failure group (2-year kidney failure risk equation $\geq 10\%$ or eGFR < 15)

Outcome	Pre-Multi-Care Kidney Clinic Eligibility Change		Post-Multi-Care Kidney Clinic Eligibility Change		Difference in Means from Post- versus Pre-Multi-Care Kidney Clinic Eligibility Change	ARIMA Model (p, d, q)	April 1, 2016, Multi-Care Kidney Clinic First Eligibility Criteria Change Point	
	Total # of Events	Mean of the Monthly Rate or Proportion (SD)	Total # of Events	Mean of the Monthly Rate or Proportion (SD)			Step Function ^a	Ramp Function ^b
							Estimate (95% CI)	Estimate (95% CI)
Multi-Care Kidney Clinic visit ^c	50,828	218.8 (20.4)	115,255	221.2 (18.3)	2.5	(0, 0, 3)	10.99 (−0.44 to 22.42)	−1.43 (−2.18 to −0.66)
Nephrologist visit ^c	95,540	413.2 (28.9)	203,721	390.8 (29.2)	−22.4	(0, 0, 0) (0, 12, 0)	1.33 (−9.07 to 11.74)	−0.59 (−1.08 to −0.11)
Dialysis initiation	3983	1.4% (0.15)	7763	1.2% (0.13)	−0.2%	(1, 0, 0)	−0.14 (−0.25 to −0.02)	−0.002 (−0.006 to 0.002)
Late referral for multidisciplinary kidney care ^d	2015	50.9% (8.2)	3123	40.3% (3.7)	−10.7%	(2, 1, 1)	−1.61 (−7.38 to 4.16)	0.70 (0.04 to 1.35)
Nondeferred dialysis starts ^d	1759	44.1% (4.0)	2987	38.5% (3.4)	−5.6%	(1, 0, 0)	−5.45 (−7.73 to −3.10)	−0.01 (−0.09 to 0.06)
Initiating dialysis with a CVC or during a hospitalization	2279	0.8% (0.13)	4373	0.7% (0.08)	−0.1%	(1, 0, 1)	−0.13 (−0.21 to 0.05)	0.000 (−0.002 to 0.003)
Home dialysis	1073	0.4% (0.06)	2193	0.3% (0.07)	−0.1%	(0, 0, 1)	0.01 (−0.03 to 0.06)	−0.002 (−0.004 to −0.000)
All-cause hospital encounters ^c	35,365	152.2 (7.7)	82,327	157.5 (5.0)	5.3	(3, 0, 0)	4.65 (−2.73 to 12.04)	−0.04 (−0.27 to 0.20)
Hospital encounters with a cardiovascular complication ^c	13,478	58.2 (4.3)	29,093	55.6 (3.6)	−2.6	(0, 0, 0) (12, 12, 0)	−1.68 (−3.44 to 0.08)	−0.01 (−0.08 to 0.05)
All-cause mortality	2226	0.8% (0.14)	5648	0.9% (0.13)	0.1%	(0, 0, 0) (12, 12, 0)	0.05 (−0.03 to 0.12)	−0.001 (−0.003 to 0.005)

ARIMA, autoregressive integrated moving average; CI, confidence interval; CVC, central venous catheter.

^aThe step coefficient represents the immediate and sustained difference in the outcome observed in the first month after the policy took effect on April 1, 2016, relative to the value that would have been expected had the preintervention trend continued unchanged.

^bThe ramp coefficient quantifies the month-to-month change in the postintervention slope compared with the preintervention slope.

^cRate per 100 person-years.

^dDenominator patients initiating dialysis.

Table 4. Difference in clinical indicators and outcomes after a change in the eligibility criteria for Multi-Care Kidney Clinics for individuals in the low-risk kidney failure group (2-year kidney failure risk equation <10% and eGFR ≥15)

Outcome	Pre-Multi-Care Kidney Clinic Eligibility Change		Post-Multi-Care Kidney Clinic Eligibility Change		Difference in Means from Post- versus Pre-Multi-Care Kidney Clinic Eligibility Change	ARIMA Model (p, d, q)	April 1, 2016, Multi-Care Kidney Clinic First Eligibility Criteria Change Point	
	Total # of Events	Mean of the Monthly Rate or Proportion (SD)	Total # of Events	Mean of the Monthly Rate or Proportion (SD)			Step Function ^a	Ramp Function ^b
							Estimate (95% CI)	Estimate (95% CI)
Multi-Care Kidney Clinic visit ^c	32,167	70.5 (7.8)	23,054	27.1 (13.7)	−43.3	(1, 1, 0) (0, 12, 0)	−1.48 (−7.55 to 4.58)	−2.10 (−4.20 to −0.00)
Nephrologist visit ^c	103,919	228.1 (26.9)	176,244	201.5 (25.9)	−26.5	(3, 0, 0) (0, 12, 0)	−6.62 (−15.26 to 2.02)	−0.50 (−0.96 to −0.03)
Dialysis initiation	529	0.1% (0.03)	761	0.1% (0.02)	0.0%	(1, 0, 0)	−0.03 (−0.04 to −0.01)	0.0002 (−0.0004 to 0.0006)
Late referral for multidisciplinary kidney care ^d	436	78.1% (10.8)	657	86.3% (8.6)	8.2%	(9, 0, 0)	3.31 (−1.46 to 8.09)	0.27 (0.12 to 0.43)
Nondeferred dialysis starts ^d	497	93.7% (6.8)	727	95.7% (5.0)	2.0%	(1, 0, 0)	3.57 (−1.41 to 8.56)	−0.04 (−0.20 to 0.11)
Initiating dialysis with a CVC or during a hospitalization	474	0.1% (0.03)	703	0.1% (0.02)	0.02%	(1, 0, 0)	−0.02 (−0.04 to −0.01)	0.000 (−0.002 to 0.003)
Home dialysis	40	0.1% (0.01)	39	0.1% (0.00)	0.0%	(3, 0, 0)	−0.003 (−0.005 to −0.001)	−0.000 (−0.000 to 0.000)
All-cause hospital encounters ^c	59,149	129.5 (6.0)	114,602	130.7 (6.2)	1.3	(0, 1, 1)	8.90 (4.70 to 13.09)	−0.34 (−0.47 to −0.20)
Hospital encounter with a cardiovascular complication ^c	19,366	42.5 (2.2)	34,709	39.6 (2.6)	−2.9	(0, 0, 0) (12, 12, 0)	−1.17 (−2.51 to 0.17)	−0.06 (−0.11 to −0.01)
All-cause mortality	3268	0.6% (0.06)	7036	0.6% (0.07)	0.1%	(3, 0, 0) (12, 12, 0)	0.04 (0.01 to 0.07)	−0.001 (−0.002 to 0.001)

ARIMA, autoregressive integrated moving average; CI, confidence interval; CVC, central venous catheter.

^aThe step coefficient represents the immediate and sustained difference in the outcome observed in the first month after the policy took effect on April 1, 2016, relative to the value that would have been expected had the preintervention trend continued unchanged.

^bThe ramp coefficient quantifies the month-to-month change in the postintervention slope compared with the preintervention slope.

^cRate per 100 person-years.

^dDenominator is patients initiating dialysis.

10,423 of 39,404 patients (27%). The redistribution of patients with low eGFR but low 2-year estimated kidney failure risk from Multi-Care Kidney Clinics to general kidney clinics resulted in more efficient resource allocation. Ontario funding for Multi-Care Kidney Clinic patients in 2015/16 (preintervention) totaled \$22 million and in 2016/17 (postintervention) totaled \$13.6 million, meaning the policy change resulted in a reduction in Multi-Care Kidney Clinic costs of \$8.4 million after the first year, which provided an opportunity to reinvest in other kidney services and more efficiently, and arguably effectively, allocate resource intensive multidisciplinary care to patients most likely to benefit from this care.²¹

Importantly, our results show that the policy change was not associated with a worsening of patient-important outcomes, such as dialysis preparation or initiation. Reassuringly, the nephrologist visit rate for the full cohort did not decline significantly; a slight, significant decline was observed for both high- and low-kidney failure risk subgroups. Given that a larger decline in the nephrologist visit rate was not observed in low-risk patients, this suggests that patients who were no longer eligible for multidisciplinary care continued to receive regular, ongoing nephrology care after the policy change.

A positive slope for late referral for multidisciplinary care (*i.e.*, less than 1 year before dialysis initiation) was observed after intervention in both high- and low-kidney failure risk subgroups. The underlying reasons for this could not be determined from these data. The fact that the increase was observed in both subgroups suggests that factors independent of the intervention, or factors tied to the intervention that affected both patients who remained Multi-Care Kidney Clinic-eligible and who became ineligible, may be contributors. As an example, some nephrologists may have remained tied to the use of eGFR for risk stratification and did not adopt the use of the KFRE into their daily clinic practice or some patients may have been missing a recent urine albumin-to-creatinine ratio measurement. Both circumstances could result in patients who are eligible for multidisciplinary care using the kidney failure risk-based criteria not being identified in a timely manner and thereby becoming a late referral for multidisciplinary care. At least 1 year of Multi-Care Kidney Clinic care is recommended by the Ontario Renal Network because late referral is associated with a higher risk of dialysis initiation during a hospitalization (on the basis of retrospective observational data).²² Reassuringly, there was no significant increase after the intervention in dialysis initiation, dialysis initiation with a CVC or in hospital, or nondeferred dialysis initiation, which are all important markers of high-quality ambulatory predialysis CKD care. The rate of dialysis initiation was also very low before and after intervention in low-risk patients (0.1%).

The underlying reasons for the significant immediate (step) increase in all-cause mortality and hospital encounters in the low-risk subgroup after the intervention are unclear. It seems unlikely that the immediate increase would be due to the intervention, given the planned phased approach of implementing the intervention over 2 years (see Methods). This assertion is further supported by the fact that there was not a significant decline in the

Multi-Care Kidney Clinic visit rate for low-risk patients immediately after the intervention (step function = -1.48 ; 95% CI, -7.55 to 4.58). Importantly, a similar pattern of positive step estimates was also observed in the high-risk group for both mortality and hospital encounters, although these estimates were not statistically significant (hospital encounters step function = 4.65 ; 95% CI, -2.73 to 12.04 ; all-cause mortality step function = 0.05 ; 95% CI, -0.03 to 0.12). This suggests that the immediate postintervention increase was not unique to low-risk patients and may reflect broader temporal or system-level factors rather than an effect specific to the eligibility change. In addition, although some step function estimates were statistically significant, the absolute effect sizes were small and unlikely to be clinically meaningful.

Our results are consistent with findings from other jurisdictions where a kidney failure risk-based approach to determine eligibility for multidisciplinary care was implemented. A study published recently using data from Alberta, Canada, also found no adverse effects from a similar policy change, but they did not apply rigorous time series analysis.²³ The Alberta study also surveyed patients and health care professionals after the policy change and found no difference in patient experience or job satisfaction.²⁴

Our study included all people in Ontario with CKD for which the intervention was applied, and we examined a balance of both patient-important outcomes and process measures. There are however important limitations to our study. The intervention was nonrandomized, so it is possible that any results observed were not directly caused by the policy but rather because of other co-occurring changes. We are however not aware of any other significant co-occurring policy or practice changes, and a randomized approach was not possible given that the Ontario Renal Network had already implemented the policy change for all centers in Ontario. Time series analysis, as used for this study, is one of the strongest nonrandomized analytic approaches to test for the effect of an intervention because it accounts for the effect of secular trends and allows for visualization of the trends before and after an intervention or policy change.²⁵ It is important to note that the observations of this study are at the population and not at the individual level; therefore, our results are ecologic in nature. Although we did not specifically adjust for potential confounders, such as changing patient demographics and risk factor profiles, the ARIMA model incorporated such factors into the background trends and tests for an effect of the change in Multi-Care Kidney Clinic eligibility criteria above and beyond these background trends.²⁶ This helps to mitigate, although does not fully eliminate, bias from unmeasured, time-varying confounders.

In conclusion, changing the eligibility criteria for multidisciplinary kidney care from an eGFR ≤ 33.4 ml/min per 1.73 m² to a 2-year kidney failure risk $\geq 10\%$ or an eGFR < 15 ml/min per 1.73 m² resulted in significantly fewer patients receiving such care without negatively affecting important clinical outcomes. We observed a slope increase in late referral for multidisciplinary care after the change, which may be an unintended consequence. Our results help to inform the implementation and administration of multidisciplinary kidney clinics in other jurisdictions.

Disclosures

Disclosure forms, as provided by each author, are available with the online version of the article at <http://links.lww.com/JSN/F762>.

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Declarative Statements

This study is exempt from the Institutional Review Board or Ethics Committee approval, and we have provided the reason for exemption. The use of data in this project was authorized under section 45 of Ontario's Personal Health Information Protection Act, which did not require review by a Research Ethics Board.

Data Availability Statements

Original data cannot be shared. Detailed Explanation Why Data Cannot Be Shared: The dataset from this study is held securely in

coded form at ICES. While legal data sharing agreements between ICES and data providers (e.g., health care organizations and government) prohibit ICES from making the dataset publicly available, access may be granted to those who meet prespecified criteria for confidential access, available at www.ices.on.ca/DAS (email: das@ices.on.ca). The full Dataset Creation Plan and underlying analytic code are available from the authors on request, understanding that the computer programs may rely on coding templates or macros that are unique to ICES and are therefore either inaccessible or may require modification.

Supplemental Material

This article contains the following supplemental material online at <http://links.lww.com/JSN/F763>.

Supplemental Figure 1. Cohort flow diagram for total number of people included in the study across all monthly intervals.

Supplemental Figure 2. Total number of people in each monthly interval after all exclusions.

Supplemental Figure 3. Proportion of people in each monthly interval followed in a Multi-Care Kidney Clinic (i.e., prevalent patients) for the overall cohort, people at high risk of kidney failure, and people at low risk of kidney failure.

Supplemental Figure 4. Proportion of patients with nondeferred dialysis starts from 2013 to 2020.

Supplemental Figure 5. Nephrologist visit rate from 2013 to 2020 (per 100 p-y).

Supplemental Figure 6. Proportion of patients initiating dialysis from 2013 to 2020.

Supplemental Figure 7. Proportion of patients initiating dialysis with a CVC or during a hospitalization from 2013 to 2020.

Supplemental Figure 8. Proportion of patients with home dialysis use from 2013 to 2020.

Supplemental Figure 9. Rate of hospital encounters with a cardiovascular complication from 2013 to 2020 (per 100 p-y).

Supplemental Table 1. Baseline characteristics for total cohort across 5 years.

Supplemental Table 2. Baseline characteristics for Multi-Care Kidney Clinic versus non-Multi-Care Kidney Clinic patients across 5 years.

Supplemental Appendix A. RECORD reporting checklist of recommendations for reporting of observational studies using the RECORD statement.

Supplemental Appendix B. Databases.

Supplemental Appendix C. Diagnostic and procedural codes for outcomes.

Supplemental Appendix D. Outcome definitions.

Supplemental Appendix E. Details on ARIMA models.

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